

Please find the detailed requirements for the Analytics Dashboard development.

Project Objective

Develop an Admin Analytics Dashboard that provides real-time and historical insights into application usage, user activity, downloads, DID creation, and uninstall statistics.

Technology Stack

Backend

- Node.js (Express.js)
- MongoDB Atlas
- Mongoose ODM
- JWT Authentication
- REST APIs

Frontend

- React.js / Next.js
- Material UI / Ant Design
- Chart.js / Recharts for analytics visualization

Database

- MongoDB Atlas

Dashboard Requirements

1. Downloads Analytics

Display download statistics with platform-wise segregation:

- Total Downloads
- Android Downloads
- iOS Downloads

Required API:

GET /api/dashboard/downloads

Expected Response:

```
{  
  "totalDownloads": 125000,  
  "androidDownloads": 92000,
```

```
"iosDownloads": 33000
}
```

2. DID Analytics

Display DID creation statistics:

- Total DIDs Created
- DIDs Created Today
- DIDs Created This Month

Required API:

GET /api/dashboard/dids

Expected Response:

```
{
"totalDids": 52341,
"todayDids": 45,
"monthlyDids": 1243
}
```

3. Uninstall Analytics

Display uninstall statistics:

- Total Uninstalls
- Android Uninstalls
- iOS Uninstalls

Required API:

GET /api/dashboard/uninstalls

Expected Response:

```
{
"totalUninstalls": 5341,
"androidUninstalls": 4321,
"iosUninstalls": 1020
}
```

User Activity Analytics

The dashboard should provide graphical representation of:

Daily Active Users (DAU)

Unique users active within a day.

API:

GET /api/dashboard/dau

Weekly Active Users (WAU)

Unique users active during a rolling 7-day period.

API:

GET /api/dashboard/wau

Monthly Active Users (MAU)

Unique users active within a month.

API:

GET /api/dashboard/mau

Dashboard UI

The dashboard should contain KPI cards displaying:

- Total Downloads
- Android Downloads
- iOS Downloads
- Total DIDs Created
- Total Uninstalls
- Active Users Today

Analytics section should include:

- DAU Trend (Line Chart)
- WAU Trend (Line Chart)
- MAU Trend (Bar Chart)

MongoDB Atlas Setup Requirements

The developer is expected to:

- Create and configure MongoDB Atlas Cluster
- Configure Database User Access
- Configure IP Whitelisting
- Create Production Database
- Create Required Collections
- Configure Atlas Connection with Node.js APIs
- Manage Environment Variables

Environment Variables:

MONGO_URI=

JWT_SECRET=

PORT=3000

Database Collections

Required collections:

1. users
2. downloads
3. uninstalls
4. user_activity

The developer should create appropriate schemas and indexing strategies to support efficient analytics queries.

Recommended Indexes:

- downloads.downloadDate
- downloads.platform
- users.createdAt
- uninstalls.uninstallDate
- uninstalls.platform
- user_activity.lastSeen
- user_activity.userId

Authentication

Implement:

- Admin Login API
- JWT Authentication
- Protected Dashboard APIs

Filtering Capabilities

The dashboard should support analytics filtering for:

- Today
- Last 7 Days
- Last 30 Days
- Last 90 Days
- Custom Date Range

Deliverables

Backend

- Complete Node.js API Application
- MongoDB Atlas Integration
- JWT Authentication
- Analytics Aggregation APIs
- Swagger Documentation

Frontend

- Responsive Dashboard UI
- KPI Cards
- DAU, WAU, MAU Graphs
- Date Filters
- Export Functionality

Deployment

- Production Deployment
- Environment Configuration
- MongoDB Atlas Production Setup
- API Health Monitoring

Estimated Timeline

The expected implementation effort is approximately 2-3 working days please see the image below for reference.

